

SHOULD HAVE A BETTER TITLE.

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ABSTRACT

We proposed a novel paradigm for implementing latent reasoning by combining DAPO-like reinforcement learning and special gating units. In this way, we transformed text-based explicit reasoning models to latent reasoning models, progressively and practically. Initial experiments demonstrate its feasibility, with significant stability and the least extra training overhead by parallelism and minimal changes, while holding full potential to scale up.

1 INTRODUCTION

The Chain of Continuous Thought (COCONUT) (Hao et al., 2024) explores the potential of the latent layer of reasoning as an alternative to the traditional explicit CoT (Wei et al., 2022) in both the reasoning procedure and the response procedure. Its previous work has shown that the latent layer of continuous reasoning can reduce unnecessary statements and focus on the core of the problem.

These models are issued from general-purpose language, trained in supervised steps of a reasoning procedure, and progressively increase their layers of latent reasoning. It is impractical due to the lack of customized reasoning data, and progressive training is significantly less convenient than explicit CoT. Hence, we provide a new approach to implementing this latent reasoning layer, fine-tuning from explicit reasoning models.

In January, the DeepSeek team proposed a distillation-based method to implement a reasoning model through reinforcement learning (DeepSeek-AI, 2025), and applied the approach to Qwen-2.5B (Yang et al., 2024a) to a reasoning model. Given the limitations of a 2.5B parameter model in reasoning tasks, we aim to implement COCONUT on the small model and find its potential in reasoning compared to the original explicit reasoning model.

2 RELATED WORK

Explicit Reasoning Models and Large-scale Reinforcement Learning Advances in explicit reasoning models, including OpenAI o1 & o3 (OpenAI, 2025; 2024) and DeepSeek R1 (DeepSeek-AI, 2025), demonstrate their strong ability in mathematical reasoning. Based on Chain-of-Thought (Wei et al., 2022), just through prompts, models achieved significant improvements on arithmetic, commonsense, and symbolic reasoning tasks with *explicitly generate* intermediate reasoning text.

Several novel paradigms were also created from explicit reasoning models (Wang et al., 2023; Yao et al., 2023; Besta et al., 2024). In this way, explicit reasoning creates a new level of mathematical reasoning tasks.

Ouyang et al. (2022) showed how GPT-3 can be fine-tuned with reinforcement learning via Proximal Policy Optimization (PPO) (Schulman et al., 2017), a reinforcement learning algorithm that optimizes the agent’s policy by maximizing the expected reward while ensuring that the new policy does not deviate too much from the old policy. Shao et al. (2024) proposed Group Relative Policy Optimization (GRPO), which is a new reinforcement learning algorithm that overcame the limitations of PPO, including the need for a large number of samples and the difficulty of tuning hyperparameters. Yu et al. (2025) demonstrated the effectiveness of Decoupled Clip and Dynamic Sampling Policy Optimization (DAPO) in large-scale reinforcement learning, which performed best in several benchmarks on Qwen2.5-32B Yang et al. (2024a) solely based on pure reinforcement learning and the base instruction model of Qwen.

Chain of Continuous Thought Previous works have already found that LLMs may involve hidden computation steps, which are not explicitly generated in text (Yang et al., 2024b; Chen et al., 2023). Driven by this spirit, Hao et al. (2024) proposed COCONUT, assigning models to “reason in a latent continuous space” through specifically designed training objectives, and demonstrated its potential compared to explicit Chain-of-Thought reasoning procedure, especially on tasks requiring backtracking. Geiping et al. (2025) introduced a new recurrent transformer architecture that lets models perform multiple hidden reasoning iterations per token. A recurrent block can be looped arbitrarily at inference to refine the hidden state before producing the next token. Chen et al. (2025) argued that LLMs may involve a born ability to reason internally, suggesting that LLMs can internally learn better reasoning policies in the latent space when given appropriate training signals. Saunshi et al. (2025) demonstrated that the critical impact of model capabilities lies in its depth instead of parameter size. In this way, they proposed *Looped Transformer*, which generated latent thoughts and simulated Chain-of-Thought reasoning procedures internally.

Compared to explicit reasoning, latent reasoning paradigms leverage latent spaces, promoting efficiency and accuracy. However, they still face problems with scaling up and stability.

3 METHODOLOGY

3.1 MODEL ARCHITECTURE

We first introduce the general structure of modern large language models (LLMs) to better demonstrate our approach and paradigm.

Modern LLMs involve Embedding, Transformer, and Output layers, as we denote them E , M_n , and O respectively, where n indicates the layer of hidden state (Vaswani et al., 2017; Radford et al., 2019).

3.2 INFERENCE

We used middle layers to enter the latent reasoning space to avoid breaking special usage in final layers (Elhage et al., 2021).

The whole inference procedure is described in (...)

We first perform prefilling by the question with the length q_0 into the model, and get the prefilled KV Cache (Pope et al., 2022) and the probability distribution of the answer’s first token t_{q_0+1} (starting to count from t_1).

After extracting the hidden state h_{n_0, q_0} (where n_0 denotes the layer number of the Transformer and q_0 denotes question length), we pass the state to a Variational Autoencoder (Kingma and Welling, 2019), and produce a vector for storage with length l_s .

Then, the new vector with the exact shape of $h_{n,k}$, as we denote $h'_{n,k}$, will be operated by several gates, as described in Equation 1.

$$h_{n,k} = h_{n,k-1} \odot g \odot \text{DMG}(e_k) \times s_{\text{inj}} + (1 - g) \odot e_k \quad (1)$$

where $e_k = \text{Embedding}(t_k)$, where t_k denotes the token of sequence with position k , and s_{inj} denotes injection scalar, which is a constant scalar to bridge the difference in order of magnitude among gating value and the result of DMG (Dynamic Current-Step-Mixing Gate).

Gates Our model has two gates: the general gate, the injection scalar, and the Dynamic Current Step Mixing Gate.

1. **General Gate.** The general gate is derived from the sequence, its value is dynamic. We denote it as g_t , and its derivation procedure can be described in Equation (not yet). We initialize it to all zeros, i.e., initially the model is equivalent to the original one, and it evolves as g moves away from 0.
2. **Dynamic Current-Step-Mixing Gate (DMG).**

Long-term Cache To enable long-term storage of reasoning features, we introduce a VAE-style bottleneck layer, compressing 1536-dimensional features to 256 dimensions and then expanding back to 1536 before gating.

We store it in each operation, and concatenate these vectors into a matrix subsequently as a hidden state.

3.3 TRAINING

We use the DAPO objective as our reinforcement learning framework, but the whole procedure is slightly different from the original DAPO.

3.3.1 REINFORCEMENT LEARNING

The whole procedure of reinforcement learning can be described in Figure.

We adapt the DAPO objective, which is formulated as:

$$\begin{aligned} \mathcal{J}_{\text{DAPO}}(\theta) = & \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \\ & \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \right] \\ \text{subject to } & 0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G, \end{aligned} \quad (2)$$

where

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \quad \hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}. \quad (3)$$

Firstly we apply rigorous datasets to the inference procedure, and verify the answer (Kydliček).

We parallelized several inference procedures simultaneously, preferably 16 for a same question, and verify the answer.

Award Mechanism We use $\text{award}_{\text{acc}}$ and $\text{penalty}_{\text{acc}}$ to award or penalize in accuracy. After mixing the accuracy and length evaluation, we inject it to loss.

Positive-Negative Ratio We use a ratio to mix positive and negative cases, which is 1 : 1.

3.4 TIME CONSISTENCY REGULARIZATION

A significant reason for chaotic response of multi-layer COCONUT is its time inconsistency, which lead to the model’s failure to maintain a consistent reasoning process.

We adapt a mechanism called “Gating Value Bonus,” which is used to guarantee the gate to converge to a fixed position, as we denote in Equation 4.

$$\mathcal{J}_{\text{gate}} = \min \left\{ C_0 \exp \left[\lambda \left(\frac{1}{2} - g_i \right)^2 \right] \right\} \quad (4)$$

3.5 PARALLELIZED TRAINING

After sampling, we concatenate the question, answer, and hidden state, and apply it to training procedure.

4 EXPERIMENTS

5 DISCUSSION

6 CONCLUSION

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A APPENDIX